Pathak Chapter 1d: Continuous Functions

Spaces of Continuous Functions, Carathéodory Functions, and Superposition Operators

Based on: An Introduction to Nonlinear Analysis and Fixed Point Theory (2018)

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What Are Continuous Functions?

Definition (Continuous Function)

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \to Y$ is **continuous** at $x_0 \in X$ if:

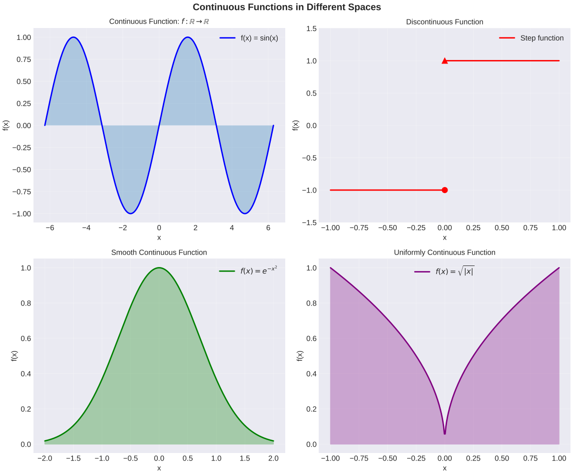
$$\forall \epsilon > 0, \exists \delta > 0 : d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \epsilon$$

f is continuous on X if it is continuous at every point of X.

Key Properties:

- Captures the intuitive notion of "no jumps"
- Preserves limits: $\lim_{n\to\infty} x_n = x \implies \lim_{n\to\infty} f(x_n) = f(x)$
- Generalizes from $\mathbb{R}$ to abstract metric and topological spaces

Continuous Functions in Different Spaces



Examples:

- Polynomial functions: $f(x) = x^2 + 1$
- Trigonometric: $f(x) = \sin(x)$
- Exponential: $f(x) = e^{-x^2}$
- Absolute value: $f(x) = |x|$ (continuous but not smooth)

Non-examples:

- Step functions (discontinuous at jumps)
- Floor function: $\lfloor x \rfloor$

The Space $C(\Omega)$

Definition (Space of Continuous Functions)

Let Ω be a metric space. The space $C(\Omega)$ consists of all continuous, real-valued functions on Ω:

$$C(\Omega) = \{f : \Omega \to \mathbb{R} \mid f \text{ is continuous}\}$$

Natural Norm (for bounded Ω):

$$\|f\|_\infty = \max_{x \in \Omega} |f(x)|$$

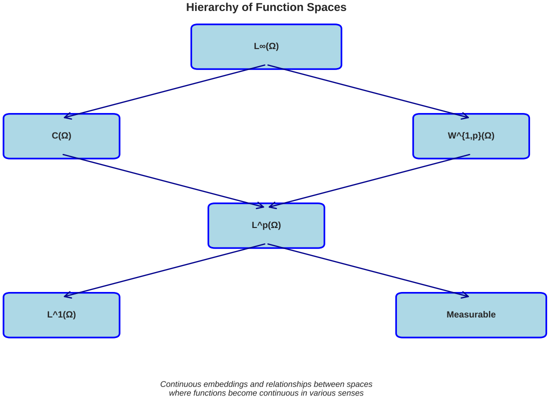
Properties of $(C(\Omega), \|\cdot\|_\infty)$:

- Complete normed space (Banach space)
- Separable if Ω is compact and separable
- Reflexive if and only if Ω is finite

Code Example: Checking Continuity in $\mathbb{R}$

```python
import numpy as np
from scipy.optimize import fminbound

def is_continuous_approx(f, a, b, x0, eps=1e-6, delta_tol=1e-4):
    """
    Approximately check continuity of f at x0 using epsilon-delta definition.
    Returns True if for given epsilon, delta exists.
    """
    # Check if |f(x) - f(x0)| < eps for all x with |x - x0| < delta
    delta = delta_tol
    x_test = np.linspace(x0 - delta, x0 + delta, 100)
    x_test = x_test[(x_test >= a) & (x_test <= b)]

    f_x0 = f(x0)
    f_x = f(x_test)
    max_diff = np.max(np.abs(f_x - f_x0))

    return max_diff < eps

# Test with f(x) = x^2 (continuous)
f = lambda x: x**2
print(is_continuous_approx(f, -1, 1, 0.5))  # True
```

Hierarchy of Function Spaces



Space Relationships:

- $C(\Omega) \subset L^\infty(\Omega)$
- $L^\infty(\Omega) \subset L^p(\Omega)$
- $W^{1,p}(\Omega)$ contains functions with weak derivatives
- All embed into spaces of measurable functions

(For bounded Ω with Lebesgue measure)

Carathéodory Functions: Definition

Definition (Carathéodory Function)

Let Ω be a measure space and U a metric space. A function $f : \Omega \times U \to \mathbb{R}$ is called a **Carathéodory function** if:

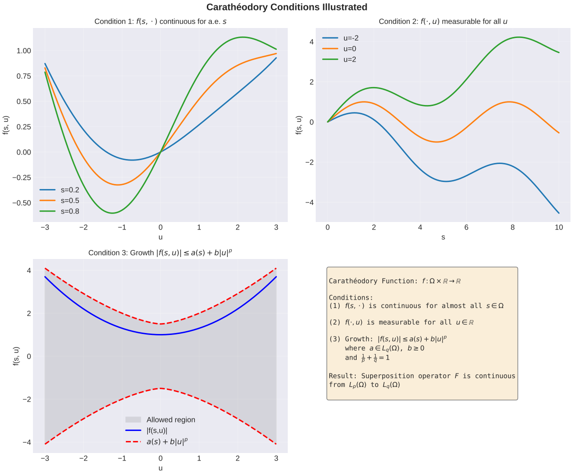
1. $f(s, \cdot)$ is **continuous** for almost all $s \in \Omega$
2. $f(\cdot, u)$ is **measurable** for all $u \in U$

Intuition:

- "Almost continuous in the second variable"
- "Measurable in the first variable"
- Natural condition for existence of solutions to differential equations

Named after: Constantin Carathéodory (1873–1950), Greek mathematician

Carathéodory Conditions in Detail



Condition 1: Continuity in u

- For each fixed s, the map $u \mapsto f(s,u)$ is continuous
- Only needs to hold for almost all s
- Allows "bad" behavior on measure-zero sets

Condition 2: Measurability in s

- For each fixed u, the map $s \mapsto f(s,u)$ is measurable
- Essential for integration theory

Growth Condition for Carathéodory Functions

Growth Condition:

$$|f(s,u)| \leq a(s) + b|u|^p$$

where:

- $a \in L_q(\Omega)$ with $q \geq 1$
- $b \geq 0$ is a constant
- $p > 0$ controls growth in u
- $\frac{1}{p} + \frac{1}{q} = 1$ (Hölder conjugates)

Significance:

- Ensures the superposition operator F maps $L_p(\Omega)$ into $L_q(\Omega)$
- Prevents too-rapid growth in the variable u
- Necessary for many existence theorems in PDE theory

Example: $f(s,u) = a(s) + b|u|$ with $a \in L_q(\Omega)$ satisfies growth with $p = 1$

Superposition Operator (Nemytski Operator)

Definition (Superposition Operator)

Let $f : \Omega \times \mathbb{R} \to \mathbb{R}$ be a function. The **superposition operator** (or **Nemytski operator**) generated by f is:

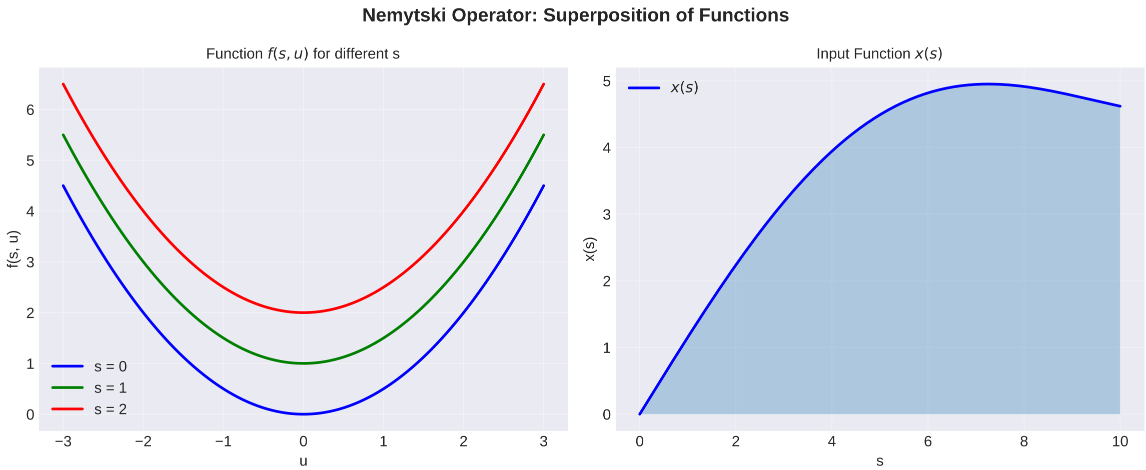
$$F(x)(s) = f(s, x(s)) \quad \text{for } s \in \Omega$$

Alternative names:

- Composition operator
- Substitution operator
- Outer superposition operator

Named after: Vladimir Nemytskii (1900–1967), Soviet mathematician

Nemytski Operator Illustrated



How it works:

1. Start with an input function $x : \Omega \to \mathbb{R}$
2. For each $s \in \Omega$, evaluate $f(s, x(s))$

Continuity of Superposition Operators

Theorem (Continuity of Nemytski Operator)

Let $f : \Omega \times \mathbb{R} \to \mathbb{R}$ satisfy the Carathéodory conditions. Then the corresponding superposition operator F is continuous and bounded from $L_p(\Omega)$ to $L_q(\Omega)$.

Key Assumption: The growth condition

$$|f(s,x)| \leq a(s) + b|x|^{p/q}$$

where $a \in L_q(\Omega)$, $b \geq 0$.

Consequence: If $\|x\|_p \to 0$, then $\|F(x)\|_q \to 0$

Code Example: Nemytski Operator

```python
import numpy as np
from scipy.integrate import quad

def nemytski_operator(f, x_func, s_vals, norm='inf'):
    """
    Compute the Nemytski operator F(x)(s) = f(s, x(s))

    Args:
        f: function f(s, u)
        x_func: input function x(s)
        s_vals: evaluation points
        norm: 'inf' for L-infinity or 'p' for L^p norm

    Returns:
        Array of F(x)(s) values
    """
    # Compute F(x)(s) = f(s, x(s)) for each s
    fx = np.array([f(s, x_func(s)) for s in s_vals])
    return fx

# Example: f(s,u) = s + u^2
f = lambda s, u: s + 0.1*u**2
x = lambda s: np.sin(s)

s = np.linspace(0, 2*np.pi, 100)
Fx = nemytski_operator(f, x, s)

```

Uniform Continuity

Definition (Uniform Continuity)

A function $f : X \to Y$ between metric spaces is **uniformly continuous** if:

$$\forall \epsilon > 0, \exists \delta > 0 : \forall x, y \in X, \quad d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \epsilon$$

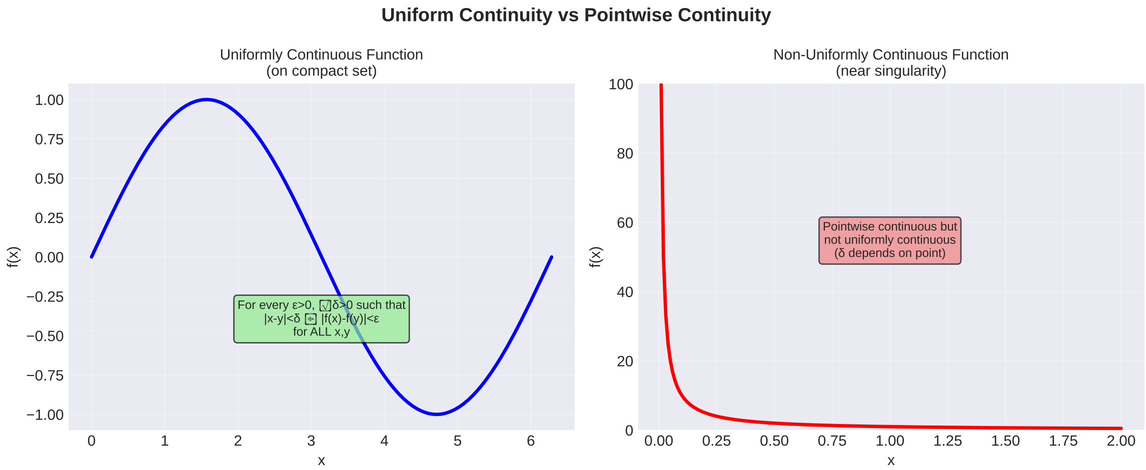
Key Difference from Continuity:

- In continuity: δ may depend on the point x_0
- In uniform continuity: δ works for *all* points simultaneously
- Much stronger condition!

Example:

- $f(x) = \sin(x)$ on $\mathbb{R}$ is uniformly continuous
- $f(x) = \frac{1}{x}$ on $(0, 1)$ is continuous but NOT uniformly continuous

Uniform Continuity Visualized



Continuous Functions on Compact Sets

Theorem (Continuous Functions on Compact Sets)

If K is a compact metric space and $f : K \to \mathbb{R}$ is continuous, then:

1. $f(K)$ *is compact*
2. f *is* **bounded***:* $\exists M > 0$ *such that* $|f(x)| \leq M$ *for all* $x \in K$
3. f **attains its extrema***:* $\exists x_{\min}, x_{\max} \in K$ *such that*

$$f(x_{\min}) = \min_{x \in K} f(x), \quad f(x_{\max}) = \max_{x \in K} f(x)$$

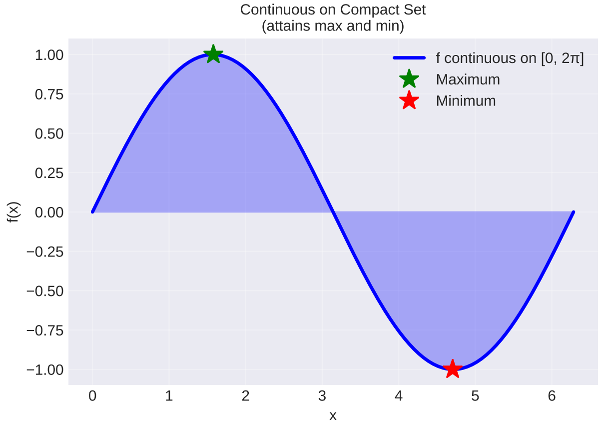
4. f *is* **uniformly continuous** *on* K

Practical Significance:

- Ensures optimization problems have solutions
- Justifies using numerical methods on compact domains

Continuous Functions on Compact Sets (Illustrated)

Continuous Functions on Compact Sets



Key Theorem: Continuous Image of Compact Set

If K is compact and $f: K \to \mathbb{R}$ is continuous, then $f(K)$ is compact.

Consequences:
- f is bounded on K
- f attains its maximum and minimum
- f is uniformly continuous on K

Example:
$f(x) = \sin(x)$ on $[0, 2\pi]$:
- Bounded: $f(x) \in [-1, 1]$
- Max: 1 (at $x = \pi/2$)
- Min: -1 (at $x = 3\pi/2$)
- Uniformly continuous

Definition (Weak Derivative)

Let $u, v \in L^p(\Omega)$ and α a multi-index. We say v is the **weak partial derivative** $D^\alpha u$ if:

$$\int_\Omega u(x) D^\alpha \varphi(x)\, dx = (-1)^{|\alpha|} \int_\Omega v(x)\varphi(x)\, dx$$

for all test functions $\varphi \in C_0^\infty(\Omega)$ (smooth with compact support).

Advantages:

- Extends differentiation beyond classical sense
- Allows handling of non-smooth functions
- Essential for modern PDE theory
- Makes spaces complete (Banach spaces)

Sobolev Spaces: Definition and Properties

Definition (Sobolev Space)

The **Sobolev space** $W^{k,p}(\Omega)$ is:

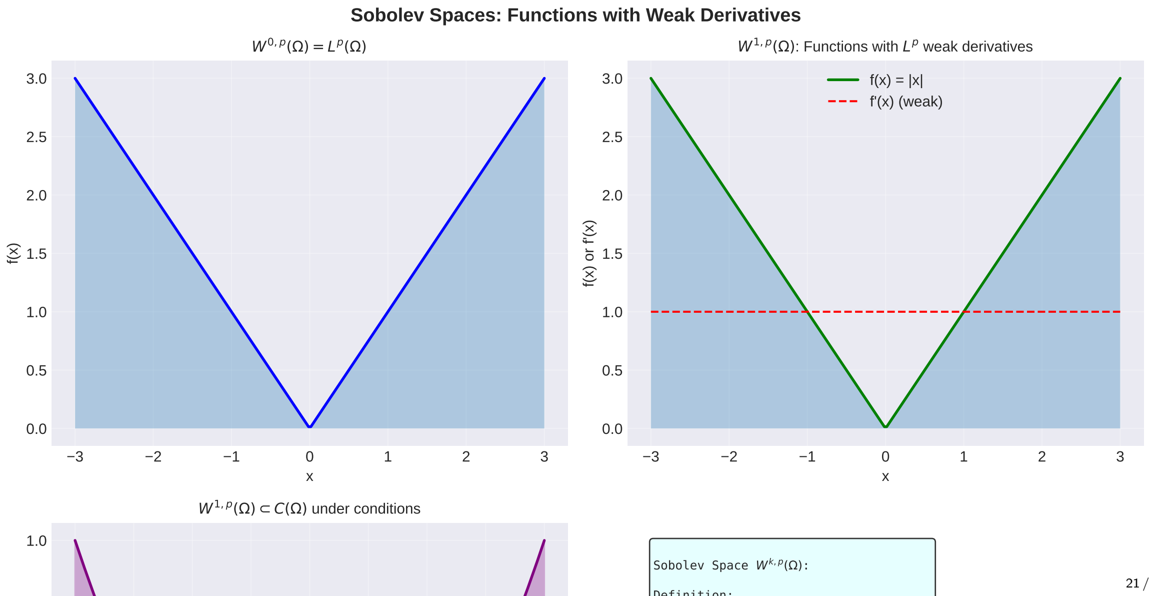
$$W^{k,p}(\Omega) = \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \text{ for all } |\alpha| \leq k\}$$

Equipped with norm:

$$\|u\|_{k,p} = \left(\sum_{|\alpha| \leq k} \int_\Omega |D^\alpha u|^p \, dx \right)^{1/p}$$

Special case $p = 2$: $H^k(\Omega) = W^{k,2}(\Omega)$ is a Hilbert space

Sobolev Spaces (Illustrated)



Sobolev Embedding Theorem

Theorem (Sobolev Embedding Theorem)

Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with smooth boundary. Then:

1. *If $0 \leq l < k$ and $\frac{1}{r} \geq \frac{1}{p} - \frac{k-l}{n}$, then*
$$W^{k,p}(\Omega) \subset W^{l,r}(\Omega)$$
with continuous embedding. The embedding is compact if $\frac{1}{r} > \frac{1}{p} - \frac{k-l}{n}$.
2. *If $\frac{1}{p} - \frac{k}{n} > 0$, then $W^{k,p}(\Omega) \subset C^{\alpha}(\Omega)$ for some $\alpha > 0$*
3. *If $\frac{1}{p} - \frac{k}{n} = 0$, then $W^{k,p}(\Omega) \subset L^{q}(\Omega)$ for all $q < \infty$*

Key takeaway: Sobolev spaces allow us to gain continuity by trading regularity for integrability.

Theorem 1.40: Carathéodory Superposition

Theorem (Theorem 1.40: Continuous Nemytski Operator)

Let $f : \Omega \times \mathbb{R} \to \mathbb{R}$ *be a Carathéodory function satisfying:*

$$|f(s,x)| \leq a(s) + b|x|^{p/q}$$

where $a \in L_q(\Omega)$, $b \geq 0$, *and* $\frac{1}{p} + \frac{1}{q} = 1$.
Then the superposition operator $F_f(x)(s) = f(s, x(s))$ *is continuous and bounded from* $L_p(\Omega)$ *to* $L_q(\Omega)$.

Application: If $x_n \to x$ in L_p, then $F(x_n) \to F(x)$ in L_q

Theorem 1.41: Superposition on L_1

Theorem (Theorem 1.41: Superposition on L_1)

Assume $f : \Omega \times \mathbb{R} \to \mathbb{R}$ satisfies the Carathéodory conditions and:

$$|f(s,x)| \leq a(s) + b|x|$$

where $a \in L_1(\Omega)$ and $b \geq 0$.
Then the superposition operator F_f maps $L_1(\Omega)$ into itself and is continuous.

Special case: Linear growth in x ensures mapping from L_1 to L_1

Theorem 1.44: Continuous Functions on Compact Sets

Theorem (Theorem 1.44: Nemytski Operator on Continuous Functions)

Let $f : \Omega \times \mathbb{R} \to \mathbb{R}$ be continuous as a map from $\Omega \times \mathbb{R} \to \mathbb{R}$, where $\Omega \subset \mathbb{R}^n$ is a closed and bounded subset.
Then the superposition operator F_f acts on $C(\Omega)$ (space of continuous functions) and is continuous and bounded.

Connection: Extends Carathéodory results to the space $C(\Omega)$

Theorem 1.49: Sobolev Embedding

Theorem (Theorem 1.49: Sobolev Embedding Theorem)

Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with smooth boundary. Then:

1. *If $0 \leq l < k$ and $\frac{1}{r} \geq \frac{1}{p} - \frac{k-l}{n}$, then*

$$W^{k,p}(\Omega) \subset W^{l,r}(\Omega)$$

   with continuous embedding (compact if strict inequality).

2. *If $\frac{1}{p} - \frac{k}{n} < 0$, then $W^{k,p}(\Omega) \subset C^{0,\alpha}(\Omega)$*

Application 1: Existence of Solutions to Integral Equations

Problem: Consider the Urysohn integral equation

$$u(s) = \int_{\Omega} K(s, t, u(t))\, dt + f(s)$$

How Continuous Functions Help:

1. View the right-hand side as a superposition operator
2. Use Carathéodory conditions on $K(s, t, u)$
3. Prove operator is continuous and compact
4. Apply Schauder fixed-point theorem

Conclusion: Equation has at least one solution

Application 2: Nonlinear Differential Equations

Problem: Existence of solutions to

$$\begin{cases} u'(t) = f(t, u(t)) \\ u(0) = u_0 \end{cases}$$

How Continuous Functions Help:

1. Rewrite as integral equation: $u(t) = u_0 + \int_0^t f(s, u(s))\, ds$
2. Carathéodory conditions ensure the integral is well-defined
3. Operator is continuous and maps appropriate space to itself
4. Apply Banach fixed-point theorem or Brouwer's theorem

Picard-Lindelöf Theorem: If f is Lipschitz in u, then unique solution exists

Application 3: Sobolev Spaces in PDE

Problem: Weak formulation of boundary value problem

$$\begin{cases} -\Delta u = f(x) & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

How Sobolev Spaces Help:

1. Seek weak solutions in $u \in H_0^1(\Omega) = W_0^{1,2}(\Omega)$
2. Weak formulation: $\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega fv \, dx$ for all test functions
3. Use Lax-Milgram theorem (requires Hilbert space structure)
4. Sobolev embedding gives regularity of solution

Result: Unique weak solution exists and belongs to $H^1(\Omega)$

Application Example: Solving an ODE Numerically

```python
import numpy as np
from scipy.integrate import odeint

# ODE: u'(t) = f(t, u(t))
def f(u, t):
    """Carath odory function f(t,u) = sin(t) * u^2"""
    return np.sin(t) * u**2

# Initial condition
u0 = 1.0

# Time grid
t = np.linspace(0, np.pi, 100)

# Solve using odeint
# Uses continuous functions internally
u_solution = odeint(f, u0, t)

# Verify: Plot solution
import matplotlib.pyplot as plt
plt.plot(t, u_solution)
plt.xlabel('t')
plt.ylabel('u(t)')
plt.title("Solution to u'(t) = sin(t)*u^2")
plt.grid(True)
plt.show()
```

Key Takeaways

- **Continuous functions** are fundamental objects preserving limits and providing natural structure for analysis
- **Carathéodory functions** extend continuity to product spaces with growth conditions
- **Superposition operators** (Nemytski) are continuous under Carathéodory conditions and growth bounds
- **Uniform continuity** on compact sets ensures boundedness and attainment of extrema
- **Sobolev spaces** extend the notion of continuity via weak derivatives, essential for modern PDE theory
- These concepts are **interdependent**: continuity $\Rightarrow$ compactness $\Rightarrow$ uniform continuity

Further Reading

From Pathak's Text:

- Chapter 1.6–1.9: Measure theory, Sobolev spaces, elliptic operators
- Chapter 3: Gâteaux and Fréchet derivatives
- Chapter 4: Monotone and maximal operators
- Chapter 5–6: Fixed point theorems and topological degree

Related References:

- *Functional Analysis* by Rudin (classical treatment)
- *Nonlinear Functional Analysis* by Deimling (applications)
- *The Sobolev Spaces in Mathematics and Their Applications*

Key Researchers Mentioned:

- Carathéodory (continuous functions)
- Nemytski (superposition operators)
- Sobolev (function spaces)
- Krasnoselskii (nonlinear operators)

Continuous Functions: The Bridge Between Analysis and Applications

Continuous functions provide the natural setting for:

- Rigorous analysis of functions and operators
- Existence and uniqueness of solutions to differential equations
- Approximation theory and numerical analysis
- Optimization and variational problems

The theory of continuous functions is **complete**, **elegant**, and **powerful**—essential for modern mathematics and applications.